

Introduction to Git and GitHub

What is Git?

- Git is a **version control system** that tracks changes in files.
- It allows you to:
 - Save versions of your work
 - Work collaboratively
 - Undo mistakes

What is GitHub?

- GitHub is a platform to **host Git repositories online**.
- Benefits:
 - Showcase your projects
 - Collaborate with others
 - Build a portfolio

1. Creating a GitHub Account

1. Go to <https://github.com> and click **Sign up**.
2. Choose a **username** (this will later be used for your personal landing page).
3. Add an email and password.
4. Confirm your email.
5. Optional: Add a profile picture and bio to personalize your account.
6. Setup 2-Factor Authentication!

2. Creating Your Landing Page (Repository with Your Username)

1. Click **New repository**.
2. **Important:** Name the repository **exactly** as your GitHub username (for example, if your username is johnsmith, the repository should be johnsmith).
3. Choose **Public** repository.
4. Check **Add a README file** (optional, but recommended).
5. Click **Create repository**.

3. Installing GitHub Desktop

1. Download GitHub Desktop: <https://desktop.github.com>
2. Install and open it.
3. Sign in with your GitHub account.

4. Working with GitHub Desktop

4.1 Cloning a Repository

1. Click **File → Clone repository**.
2. Select your landing page repository (username.github.io).
3. Choose a local folder.

4.2 Adding Files

1. Open the folder in **GitHub Desktop** or your file explorer.
2. Add files (e.g., README.md or a simple HTML file).

4.3 Committing Changes

1. Open GitHub Desktop → You will see **Uncommitted Changes**.
2. Add a **commit message** (e.g., "Update bio").
3. Click **Commit to main**.

4.4 Pushing to GitHub

1. Click **Push origin** in GitHub Desktop.
2. Your changes are now live on GitHub.

4.5 Pulling Updates

1. Click **Fetch origin → Pull** if changes exist online.
2. Keep your local repo updated.

5. Exercises

5.1 Exercise 1: Personal Landing Page

1. Clone your username repository using GitHub Desktop.
2. Edit README.md with:
 - Your name
 - Short bio
 - Skills or hobbies
3. Commit and push changes.
4. Verify your page at: <https://<username>.github.io>
5. Hint: You can use <https://profile-readme-generator.com/> to create an awesome page

5.2 Exercise 2: Git Basics with GitHub Desktop

1. Create a new repository in GitHub Desktop called PYTWO.
2. Add a file README.md with a short description.
3. Commit and push the changes.
4. Check your repository on GitHub.
5. This should be the repository that you use in PYTWO.